

Question 1

question1.cpp ×

question2.cpp

question3.cpp

C++

question1.cpp > main()

```
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cin >> n;
6      int count = 0;
7      for (int i=0; n>0; ) {
8          count++ ;
9          n = n/10;
10     }
11     cout << count ;
12 }
```

PROBLEMS

OUTPUT

TERMINAL

Code +

|

|

|

|

```
cd "/Users/chirag/Documents/ass4/" && g++ question1.cpp -o
question1 && "/Users/chirag/Documents/ass4/"question1
● Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4
/" && g++ question1.cpp -o question1 && "/Users/chirag/Doc
uments/ass4/"question1
12345
5%
○ Chirag ~/Documents/ass4 %
```

I

Question 2

```
question2.cpp > main()
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cin >> n;
6      int ans = 0;
7      while (n>0) {
8          int digit = n%10;
9          ans = ans*10 + digit;
10         n = n/10;
11     }
12     cout << ans ;
13 }
```

```
Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question2.cpp -o question2 && "/Users/chirag/Documents/ass4/"question2
12345
54321%
Chirag ~/Documents/ass4 %
```

I



Ln 13, Col 2

Spaces: 4

UTF-8

LF

{ } C++



Go Live

Mac

Prettier



Question 3

```
question3.cpp > main()
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cin >> n;
6      int ans = 0;
7      while(n>0) {
8          int digit = n%10 ;
9          ans += digit ;
10         n = n/10 ;
11     }
12     cout << ans;
13 }
```

Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question3.cpp -o question3 && "/Users/chirag/Documents/ass4/"question3
12345
15%

Chirag ~/Documents/ass4 %

Question 4

PROBLEMS

TERMINAL

Code + -

```
● Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question4.cpp -o question4 && "/Users/chirag/Documents/ass4/"question4
123456
3 3%
○ Chirag ~/Documents/ass4 %
```

```
question4.cpp > main()
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cin >> n;
6      int even = 0;
7      int odd = 0;
8      while(n>0) {
9          int digit = n%10 ;
10         if (digit%2 == 0) {
11             even++;
12         }
13         else {
14             odd++;
15         }
16         n = n/10 ;
17     }
18     cout << even << " " << odd;
19 }
```


Question 5

PROBLEMS

TERMINAL

Code + -

```
● Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question5.cpp -o question5 && "/Users/chirag/Documents/ass4/"question5
472953
9%
○ Chirag ~/Documents/ass4 %
```

```
question5.cpp > main()
1  #include <iostream>
2  using namespace std;
3  int main() {
4      int n;
5      cin >> n;
6      int ans = 0;
7      while(n>0) {
8          int digit = n%10 ;
9          if (digit>ans) {
10             ans = digit;
11         }
12         n = n/10 ;
13     }
14     cout << ans;
15 }
```

Question 6

question6.cpp > main()

```
2  using namespace std;
3  int main(){
4      int l, r ;
5      cin >> l >> r ;
6      int cnt = 0;
7      if(l==r || r<l || r==1) {
8          cout << "0";
9      }
10     else {
11         int n = l;
12         while(n<r) {
13             int isPrime = 1;
14             for (int i=2; i<n; i++) {
15                 if(n%i==0) {
16                     isPrime = 0;
17                 }
18             }
19             if(isPrime) {
20                 cnt++;
21             }
22             n++;
23         }
24         cout << cnt ;
25     }
26 }
```

Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question6.cpp -o question6 && "/Users/chirag/Documents/ass4/"question6
11 20
4%
Chirag ~/Documents/ass4 %

Question 7

PROBLEMS

TERMINAL

Code + -

```
● Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question7.cpp -o question7 && "/Users/chirag/Documents/ass4/"question7
12
16%
○ Chirag ~/Documents/ass4 %
```

```
question7.cpp > main()
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4  int main() {
5      int n;
6      cin >> n;
7      vector<int> ans = {0};
8      int i = 1;
9      int sum = 0;
10     while(i<n) {
11         for (int j=1; j<=n; j++) {
12             if( (i*j) == n ) {
13                 ans.push_back(i);
14             }
15         }
16         i++;
17     }
18     for(int i=0; i<ans.size(); i++) {
19         sum += ans[i] ;
20     }
21     cout << sum ;
22
23 }
24
```


Question 8

PROBLEMS

TERMINAL

Code + -

```
● Chirag ~/Documents/ass4 % cd "/Users/chirag/Documents/ass4/" && g++ question8.cpp -o question8 && "/Users/chirag/Documents/ass4/"question8
12
1 2 3 4 6 12 %
○ Chirag ~/Documents/ass4 %
```

```
question8.cpp > main()
1  #include <iostream>
2  #include <vector>
3  using namespace std;
4  int main() {
5      int n;
6      cin >> n;
7      vector<int> ans;
8      int i = 1;
9      while(i<=n) {
10         for (int j=1; j<=n; j++) {
11             if( (i*j) == n ) {
12                 ans.push_back(i);
13             }
14         }
15         i++;
16     }
17     sort(ans.begin(), ans.end()) ;
18     for(int i=0; i<ans.size(); i++) {
19         cout << ans[i] << " " ;
20     }
21 }
```